

48. LOAD COMB 1001 1.4DL + 1.6LL
49. 1 1.4 2 1.4 3 1.6
50. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ
51. 1 1.2 2 1.2 3 0.5 21 1.0
52. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ
53. 1 1.2 2 1.2 3 0.5 21 -1.0
54. LOAD COMB 1221 0.9DL + 1.0EQ
55. 1 0.9 2 0.9 21 1.0
56. LOAD COMB 1321 0.9DL - 1.0EQ
57. 1 0.9 2 0.9 21 -1.0
59. LOAD COMB 2001 1.0DL + 1.0LL
60. 1 1.0 2 1.0 3 1.0
61. LOAD COMB 2002 1.0DL + 0.5LL
62. 1 1.0 2 1.0 3 0.5
64. PERFORM ANALYSIS

65. PRINT MAXFORCE ENV

MAXFORCE ENV

JA B18 FRAME 1

3

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBERS/ELEMENTS/SUPPORTS = 6/ 6/ 2
ORIGINAL/FINAL BAND-WIDTH= 3/ 3/
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 14
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQD/AVAIL. DISK SPACE = 12.0/ 16025.1 MB

65. PRINT MAXFORCE ENV

MAXFORCE ENV

JA B18 FRAME 1

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--- PAGE NO.

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MAX AND MIN FORCE VALUES AMONGST ALL SECTION LOCATIONS

MEMB	FY/	DIST	ID	MZ/	DIST	ID	FX	DIST
	FZ	DIST	LD	MY	DIST	LD		
101 MAX	55.22	0.00	1321	549.98	1.60	1021	1244.85	C 0.00
	0.00	0.00	1	0.00	0.00	1		
MIN	-343.73	1.60	1021	-88.35	1.60	1321	95.56	C 1.60
	0.00	1.60	2002	0.00	1.60	2002		
201 MAX	118.27	0.00	1321	355.25	6.41	1021	350.15	C 0.00
	0.00	0.00	1	0.00	0.00	1		
MIN	-207.75	0.00	1021	-483.35	0.00	1021	19.66	C 6.41
	0.00	6.41	2002	0.00	6.41	2002		
102 MAX	343.73	0.00	1121	246.14	1.60	21	1244.85	C 0.00
	0.00	0.00	1	0.00	0.00	1		
MIN	-153.84	1.60	21	-549.98	1.60	1121	189.54	T 1.60
	0.00	1.60	2002	0.00	1.60	2002		

2

STAAD.Pro
Version 2006 Bld 1005.US
Proprietary Program of
Research Engineers, Intl.
Date= OCT 29, 2007
Time= 18:23: 8

USER ID: Meinhardt Infrastructure Pte Ltd

1. STAAD PLANE JA B18 FRAME 1

INPUT FILE: JA B18 RC Frame-1.STD

2. START JOB INFORMATION

3. ENGINEER DATE 29-OCT-07

4. END JOB INFORMATION

5. INPUT WIDTH 79

6. UNIT METER KN

7. JOINT COORDINATES

8. 101 0 0 0

9. 201 0 1.6 0

10. 301 0 8.01 0

11. 102 7.8 0 0

12. 202 7.8 1.6 0

13. 302 7.8 8.01 0

14. MEMBER INCIDENCES

15. 101 101 201

16. 201 201 301

17. 102 102 202

18. 202 202 302

19. 2001 201 202

20. 3001 301 302

21. MEMBER PROPERTY

22. 101 TO 102 PRI ZD 0.55 YD 0.55

23. 201 TO 202 PRI ZD 0.5 YD 0.5

24. 2001 PRI ZD 0.4 YD 1

25. 3001 PRI ZD 0.3 YD 0.7

26. CONSTANTS

27. E 2.5E+007 MEMB 101 TO 102 201 TO 202 2001 3001

28. POISSON 0.2 MEMB 101 TO 102 201 TO 202 2001 3001

29. DENSITY 24 MEMB 101 TO 102 201 TO 202 2001 3001

30. SUPPORTS

31. 101 102 PIN

32. LOAD 1 SELFWEIGHT

33. SELFWEIGHT Y -1

34. LOAD 2 SDL

35. MEMBER LOAD

36. 2001 UNI GY -69

37. 3001 UNI GY -41

38. LOAD 3 LL

39. MEMBER LOAD

40. 2001 UNI GY -72

JA B18 FRAME 1

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--- PAGE NO.

41. 3001 UNI GY -7.2

43. LOAD 21 EQ (EARTHQUAKE LOAD)

44. MEMBER LOAD

45. 201 UNI Y 24

46. 202 UNI Y 24

1

21

202 MAX 207.75 0.00 1121 483.35 0.00 1121 350.15 C 0.00
1001 0.00 0.00 1
MIN -153.84 0.00 21 -355.25 6.41 1121
0.00 6.41 2002 41.73 T 6.41
21

2001 MAX 878.44 0.00 1001 1033.32 0.00 1021 264.22 C 0.00
0.00 0.00 1
MIN -878.44 7.80 1001 -937.37 3.90 1001
0.00 7.80 2002 0.00 7.80
21

3001 MAX 296.31 0.00 1001 355.25 0.00 1021 76.08 C 0.00
0.00 0.00 1
MIN -296.31 7.80 1001 -321.26 3.90 1001
0.00 7.80 2002 0.00 7.80
21

***** END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

67. LOAD LIST ALL

68. PRINT SUPPORT REACTIONS ALL

□ SUPPORT REACTION ALL

JA B18 FRAME 1

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--- PAGE NO.

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT LOAD FORCE-X FORCE-Y FORCE-Z MOM-X MOM-Y MOM-Z

101 1 13.66 107.17 0.00 0.00 0.00 0.00
2 95.92 429.00 0.00 0.00 0.00 0.00
3 116.80 308.88 0.00 0.00 0.00 0.00
21 153.84 189.54 0.00 0.00 0.00 0.00
1001 340.29 1244.85 0.00 0.00 0.00 0.00
1021 343.73 987.39 0.00 0.00 0.00 0.00
1121 36.05 608.31 0.00 0.00 0.00 0.00
1221 252.46 672.09 0.00 0.00 0.00 0.00
1321 -55.22 293.02 0.00 0.00 0.00 0.00
2001 226.38 845.05 0.00 0.00 0.00 0.00
2002 167.98 690.61 0.00 0.00 0.00 0.00
102 1 -13.66 107.17 0.00 0.00 0.00 0.00
2 -95.92 429.00 0.00 0.00 0.00 0.00
3 -116.80 308.88 0.00 0.00 0.00 0.00
21 153.84 -189.54 0.00 0.00 0.00 0.00
1001 -340.29 1244.85 0.00 0.00 0.00 0.00
1021 -343.73 987.39 0.00 0.00 0.00 0.00
1121 -36.05 608.31 0.00 0.00 0.00 0.00
1221 -252.46 672.09 0.00 0.00 0.00 0.00
1321 -55.22 293.02 0.00 0.00 0.00 0.00
2001 -226.38 845.05 0.00 0.00 0.00 0.00
2002 -167.98 690.61 0.00 0.00 0.00 0.00

***** END OF LATEST ANALYSIS RESULT *****

69. PRINT MEMBER FORCES ALL

□ MEMBER FORCES ALL

JA B18 FRAME 1

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--- PAGE NO.

MEMBER END FORCES STRUCTURE TYPE = PLANE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER LOAD JT AXIAL SHEAR-Y SHEAR-Z TORSTION MOM-Z

101 1 101 107.17 -13.66 0.00 0.00
0.00 201 -95.56 13.66 0.00 0.00
2 101 429.00 -95.92 0.00 0.00
0.00 201 -429.00 95.92 0.00 0.00
3 101 308.88 -116.80 0.00 0.00
0.00 201 -308.88 116.80 0.00 0.00
21 101 189.54 -153.84 0.00 0.00
0.00 201 -189.54 153.84 0.00 0.00
1001 101 1244.85 -340.29 0.00 0.00
0.00 201 -1228.59 340.29 0.00 0.00
1021 101 987.39 -343.73 0.00 0.00
0.00 201 -973.45 343.73 0.00 0.00
1121 101 608.31 -36.05 0.00 0.00
0.00 201 -594.37 36.05 0.00 0.00
1221 101 672.09 -252.46 0.00 0.00
0.00 201 -661.64 252.46 0.00 0.00
1321 101 293.02 55.22 0.00 0.00
0.00 201 -282.56 -55.22 0.00 0.00
2001 101 845.05 -226.38 0.00 0.00
0.00 201 -833.44 226.38 0.00 0.00
2002 101 690.61 -167.98 0.00 0.00
0.00 201 -679.00 167.98 0.00 0.00
201 1 201 58.12 -4.44 0.00 0.00
-12.19 301 -19.66 4.44 0.00 0.00
2 201 159.90 -35.08 0.00 0.00